12 Discrete Mathematics: Set Theory & Algebra (20)

12.0.1 UGC NET CSE | July 2016 | Part 2 | Question: 1

How many difference equivalence relations with exactly three different equivalence classes are there on a set with five elements?

A. 10 B. 15 C. 25 D. 30

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Answer key ☟

12.0.2 UGC NET CSE | June 2019 | Part 2 | Question: 9

Find the zero-one matrix of the transitive closure of the relation given by the matrix A :

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

A. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

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Answer key ☟

12.0.3 UGC NET CSE | December 2004 | Part 2 | Question: 2

If $f(x) = x + 1$ and $g(x) = x + 3$ then $f0f0f0f$ is :

A. g B. $g+1$ C. g^4 D. None of these

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Answer key ☟

12.1 Boolean (1)

12.1.1 Boolean: UGC NET CSE | June 2019 | Part 2 | Question: 13

How many different Boolean functions of degree n are the

A. 2^{2^n} B. $(2^2)^n$ C. $2^{2^n} - 1$ D. 2^n

ugcnetcse-june2019-paper2 boolean function

Answer key ☟

12.2 Complement In Fuzzy Set (1)

12.2.1 Complement In Fuzzy Set: UGC NET CSE | June 2014 | Part 3 | Question: 07

Given

$U = \{1, 2, 3, 4, 5, 6, 7\}$

$A = \{(3, 0.7), (5, 1), (6, 0.8)\}$

then $\tilde{A}$ will be: (where $\sim\rightarrow$ complement)

A. $\{(4, 0.7), (2, 1), (1, 0.8)\}$
B. $\{(4, 0.3), (5, 0), (6, 0.2)\}$
C. $\{(1, 1), (2, 1), (3, 0.3), (4, 1), (6, 0.2), (7, 1)\}$
D. $\{(3, 0.3), (6, 0.2)\}$

ugcnetjune2014iii fuzzy-set complement-in-fuzzy-set

Answer key

12.3 Euler Phi Function (1)

12.3.1 Euler Phi Function: UGC NET CSE | June 2019 | Part 2 | Question: 70

Consider the following statements:

S_1: For any integer $n > 1$, $a^{\phi(n)} \equiv 1(mod\, n)$ for all $a \in Z_n^*$, where $\phi(n)$ is Euler's phi function.

S_2: If p is prime, then $a^p \equiv 1(mod\, p)$ for all $a \in Z_p^*$.

Which one of the following is/are correct?

A. Only S_1
B. Only S_2
C. Both S_1 and S_2
D. Neither S_1 nor S_2

ugcnetcse-june2019-paper2 euler-phi-function set-theory&algebra

Answer key

12.4 Factors (1)

12.4.1 Factors: UGC NET CSE | January 2017 | Part 2 | Question: 4

How many multiples of 6 are there between the following pairs of numbers?

0 and 100 and -6 and 34

A. 1 and 6
B. 17 and 6
C. 17 and 7
D. 16 and 7

ugcnetjan2017ii set-theory&algebra factors

Answer key

12.5 Function Composition (1)

12.5.1 Function Composition: UGC NET CSE | December 2013 | Part 2 | Question: 37

Let f and g be the functions from the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Then the composition of f and g and g and f is given as

A. 6x+7, 6x+11
B. 6x+11, 6x+7
C. 5x+5, 5x+5
D. None of the above

ugcnetcse-dec2013-paper2 algebra function-composition

Answer key

12.6 Functions (3)

12.6.1 Functions: UGC NET CSE | December 2014 | Part 2 | Question: 05

If we define the functions f, g and h that map R into R by :

$f(x) = x^4, g(x) = \sqrt{x^2 + 1}, h(x) = x^2 + 72$, then the value of the composite functions $ho(gof)$ and $(hog)of$ are given as

A. $x^8 - 71$ and $x^8 - 71$
B. $x^8 - 73$ and $x^8 - 73$
C. $x^8 + 71$ and $x^8 + 71$
D. $x^8 + 73$ and $x^8 + 73$

ugcnetcse-dec2014-paper2 discrete-mathematics functions

Answer key

12.6.2 Functions: UGC NET CSE | January 2017 | Part 2 | Question: 3

The functions mapping R into R are defined as :
$f(x) = x^3 - 4x, g(x) = \frac{1}{x^2+1}$ and $h(x) = x^4$.
Then find the value of the following composite functions :
$h_o g(x)$ and $h_o g_o f(x)$

A. $\left(x^2+1\right)^4$ and $\left[\left(x^3-4x\right)^2+1\right]^4$
B. $\left(x^2+1\right)^4$ and $\left[\left(x^3-4x\right)^2+1\right]^{-4}$
C. $\left(x^2+1\right)^{-4}$ and $\left[\left(x^2-4x\right)^2+1\right]^4$
D. $\left(x^2+1\right)^{-4}$ and $\left[\left(x^3-4x\right)^2+1\right]^{-4}$

ugcnetjan2017ii functions set-theory&algebra

Answer key

12.6.3 Functions: UGC NET CSE | September 2013 | Part 2 | Question: 25

If F and G are Boolean functions of degree n. Then, which of the following is true?

A. F $\leq$ F + G and F G $\leq$ F
B. G $\leq$ F + G and F G $\geq$ G
C. F $\geq$ F + G and F G $\leq$ F
D. G $\geq$ F + G and F G $\leq$ F

functions ugcnetsep2013ii

Answer key

12.7 Inclusion Exclusion (1)

12.7.1 Inclusion Exclusion: UGC NET CSE | October 2020 | Part 2 | Question: 1

The number of positive integers not exceeding 100 that are either odd or the square of an integer is _______

A. 63
B. 59
C. 55
D. 50

ugcnetcse-oct2020-paper2 discrete-mathematics inclusion-exclusion

Answer key

12.8 Poset (1)

12.8.1 Poset: UGC NET CSE | June 2019 | Part 2 | Question: 1

Consider the poset $(\{3,5,9,15,24,45\},|)$.
Which of the following is correct for the given poset ?

A. There exist a greatest element and a least element
B. There exist a greatest element but not a least element
C. There exist a least element but not a greatest element
D. There does not exist a greatest element and a least element

ugcnetcse-june2019-paper2 poset set-theory&algebra

Answer key

12.9 Power Set (1)

12.9.1 Power Set: UGC NET CSE | December 2012 | Part 2 | Question: 4

The power set of the set $\{\Phi\}$ is

A. $\{\Phi\}$
B. $\{\Phi, \{\Phi\}\}$
C. $\{0\}$
D. $\{0, \Phi, \{\Phi\}\}$

ugcnetcse-dec2012-paper2 set-theory&algebra set-theory power-set

Answer key

12.10 Relational Algebra (1)

12.10.1 Relational Algebra: UGC NET CSE | December 2008 | Part 2 | Question: 12

A relation R in $\{1,2,3,4,5,6\}$ is given by $\{(1,2),(2,3),(3,4),(4,4),(4,5)\}$. This relation is:

A. reflexive
B. symmetric
C. transitive
D. not reflexive, not symmetric and not transitive

ugcnetcse-dec2008-paper2 set-theory&algebra relational-algebra

Answer key

12.11 Relations (5)

12.11.1 Relations: UGC NET CSE | December 2009 | Part 2 | Question: 01

If she is my friend and you are her friend, then we are friends. Given this, the friend relationship in this context is ____________.

(i) commutative (ii) transitive (iii) implicative (iv) equivalence

(A) (i) and (ii)

(B) (iii)

(C) (i), (ii), (iii) and (iv)

(D) None of these

ugcnetcse-dec2009-paper2 set-theory&algebra relations

Answer key

12.11.2 Relations: UGC NET CSE | July 2016 | Part 2 | Question: 3

Suppose that R_1 and R_2 are reflecive relations on a set A. Which of the following statements is correct?

A. $R_1 \cap R_2$ is reflexive and $R_1 \cup R_2$ is irreflexive
B. $R_1 \cap R_2$ is irreflexive and $R_1 \cup R_2$ is reflexive
C. Both $R_1 \cap R_2$ and $R_1 \cup R_2$ are reflexive
D. Both $R_1 \cap R_2$ and $R_1 \cup R_2$ are irreflexive

ugcnetcse-july2016-paper2 discrete-mathematics relations

Answer key

12.11.3 Relations: UGC NET CSE | June 2005 | Part 2 | Question: 3

The transitive closure of a relation R on set A whose relation matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ is :

A. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

ugcnetcse-june2005-paper2 discrete-mathematics relations

Answer key

12.11.4 Relations: UGC NET CSE | June 2010 | Part 2 | Question: 1

"x^1 is a clone of x" means x^1 is identical to x in terms of the physical attributes namely, height, weight and complexion. Given, height, weight and complexion only form a complete set of attributes for an entity, cloning is an equivalence relation. What is your impression about this statement ?

A. The statement is true
B. The statement is false
C. The truth value of the statement cannot be computed
D. None of these

ugcnetcse-june2010-paper2 set-theory&algebra relations

Answer key

12.11.5 Relations: UGC NET CSE | June 2013 | Part 2 | Question: 34

The relation “devides” on a set of positive integers is_____

A. Symmetric and transitive
B. Anti symmetric and transitive
C. Symmetric only
D. Transitive only

ugcnetcse-june2013-paper2 relations set-theory&algebra

Answer key

Answer Keys

12.0.1	C	12.0.2	A	12.0.3	Q-Q	12.1.1	A	12.2.1	C
12.3.1	A	12.4.1	C	12.5.1	A	12.6.1	D	12.6.2	D
12.6.3	A	12.7.1	C	12.8.1	D	12.9.1	B	12.10.1	Q-Q
12.11.1	Q-Q	12.11.2	C	12.11.3	Q-Q	12.11.4	Q-Q	12.11.5	B